# ArriBot

AI-Powered Holographic Learning Companion

COMPREHENSIVE PRODUCT REQUIREMENTS DOCUMENT (PRD)

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# 1. Product Vision & Mission

### Product Vision

To be the ultimate digital companion for lifelong learners, replacing static textbooks, manual flashcards, and disjointed tools with a unified, intelligent, and immersive platform that actively adapts to each user's career and learning goals.

We envision a future where learning is not a chore but an engaging, gamified experience driven by personalized AI.

### Mission Statement

To democratize access to high-quality, personalized education and career coaching by removing the friction of content creation. ArriBot aims to provide an engaging, beautifully designed spatial environment that fosters deep focus, continuous growth, and rapid skill acquisition.

### Core Value Proposition

- **For Students:** Instant generation of personalized study materials (flashcards, quizzes, summaries) from any topic or text, eliminating hours of manual overhead.
- **For Job Seekers:** Deep, AI-driven resume gap analysis aligned directly with targeted career roles, instantly generating learning paths to bridge those gaps.
- **For All Learners:** An inspiring, gamified, "spaceship command center" interface that turns studying into an engaging daily habit.

## 2. Problem Statement

### The Current Landscape

The EdTech market is saturated but highly fragmented. Learners are forced to string together multiple disconnected applications:

- **Flashcard tools** (Anki, Quizlet) for rote memorization.
- **Generic AI chatbots** (ChatGPT, Claude) for unstructured Q&A and tutoring.
- **Static curriculum platforms** (Roadmap.sh, Coursera) for learning paths.
- **Expensive resume review services** (TopResume) for career transitions.

### The Core Pain Points

1. **The Content Creation Bottleneck:** Students spend up to 40% of their study time preparing to study—typing out flashcards, structuring notes, and searching for practice questions.
2. **The "Blank Canvas" Problem of GenAI:** While generic chatbots are powerful, users often lack structured progression, daily milestones, and gamified tracking of their learning journey. Chats are ephemeral and unstructured.
3. **Disconnected Career Prep:** Job seekers are told what to learn for a specific job but are left on their own to figure out how to learn those things and in what order.
4. **Low Engagement & High Churn:** Traditional Learning Management Systems (LMS) and study apps suffer from visually sterile, spreadsheet-like interfaces. This leads to rapid motivation drop-off.

## 3. Target Users & Personas

### Persona 1: "Overwhelmed Olivia" (The University Student)

- **Profile:** 20 years old, studying Computer Science or Pre-Med, juggling 5 dense courses simultaneously.
- **Goals:** Memorize vast amounts of terminology; prepare for midterms efficiently; understand complex concepts quickly.
- **Frustrations:** Creating Anki decks takes hours. Lecture slides are too dense to read the night before an exam.
- **ArriBot Solution:** Uses *Flash Recall* to instantly turn a syllabus topic into flashcards and the *Summarizer* to compress dense lecture PDFs into 1-page key takeaways.

### Persona 2: "Self-Taught Sam" (The Aspiring Developer)

- **Profile:** 25 years old, working a non-tech job, learning coding via scattered YouTube tutorials and blogs.
- **Goals:** Build a portfolio, master a tech stack, land a junior developer role.
- **Frustrations:** Suffers from "tutorial hell." Doesn't have a structured curriculum.
- **ArriBot Solution:** Uses the *Learning Path Generator*. Sam inputs "Master React in 8 weeks," and ArriBot generates a day-by-day checklist.

### Persona 3: "Transitioning Taylor" (The Job Seeker)

- **Profile:** 28 years old, moving from QA Engineering to Backend Engineering.
- **Goals:** Optimize resume for ATS; identify missing skills; prep for technical interviews.
- **Frustrations:** Unsure what specific backend concepts (e.g., Docker, Kubernetes) they lack.
- **ArriBot Solution:** Uploads resume to the *Resume Analyzer*, enters target role "Backend Engineer". ArriBot highlights missing skills and generates a learning path to bridge the gap.

## 4. Competitive Analysis

| Competitor | Strengths | Weaknesses | ArriBot's Advantage |
|---|---|---|---|
| **Quizlet** | Huge existing user base, community decks. | Manual creation is slow, limited AI features, paywalls on basic functions. | Instant AI generation of cards from raw text/PDFs. |
| **ChatGPT / Claude** | Highly intelligent, versatile. | No structured learning paths, no spaced repetition, no gamification. | Purpose-built UI for studying, integrated curriculum tracking, focused workflows. |
| **Anki** | Best-in-class spaced repetition algorithm. | Extremely steep learning curve, outdated UI, tedious card creation. | Visually stunning UI, frictionless AI card creation. |
| **Coursera / Udemy** | Accredited courses, high-quality video content. | Static, one-size-fits-all curriculum, cannot adapt to specific resume gaps. | Hyper-personalized learning paths generated in seconds based on a user's exact needs. |

## 5. Product Requirements Document (PRD)

This section outlines the core epics and functional requirements of the system.

### Epic 1: Identity, Authentication & Security

- **REQ-1.1:** Users register with an email and password. OTP verification via email (Brevo) within 10 minutes.
- **REQ-1.2:** Session management using JSON Web Tokens (JWT) signed with HS256, expiring in 24 hours.
- **REQ-1.3:** Protected routes on the React frontend redirecting unauthenticated users to `/auth`.

### Epic 2: The Core AI Modules

- **REQ-2.1 — Neural Chat:** AI tutor interface with markdown support, code highlighting, and chat history.
- **REQ-2.2 — Flash Recall:** Instantly generate 1-20 3D flippable flashcards from a topic or text.
- **REQ-2.3 — Cognitive Quiz:** Generate 4-option multiple choice quizzes with instant grading and explanations.
- **REQ-2.4 — Summarizer:** Compress up to 10,000 characters of text into TL;DR, Bullet Points, and Key Takeaways.
- **REQ-2.5 — Resume Analyzer:** Upload PDF resume, input target role, and receive a Suitability Score, missing skills list, and customized learning path.
- **REQ-2.6 — Learning Path Generator:** Create nested, daily curricula for any macro-goal (e.g., "Learn Python in 4 weeks").

### Epic 3: Resource Quotas & Usage Tracking

- **REQ-3.1:** Implement daily feature limits (e.g., 30 chats, 10 flashcard sets).
- **REQ-3.2:** Sliding window rate limiter (Max 5 requests per minute per user).
- **REQ-3.3:** Track AI prompt and completion tokens per user in MongoDB.

### Epic 4: UX & Interface Design

- **REQ-4.1:** Implement a "Spaceship Command Center" aesthetic (Deep void background, neon accents, glassmorphism).
- **REQ-4.2:** Fluid animations and micro-interactions (Framer Motion) for all components.

## 6. User Stories

### Authentication & Onboarding

- As a new user, I want to sign up securely with OTP so that my account is protected.

- As a returning user, I want to remain logged in via JWT so I don't have to authenticate every session.

### Learning & Content Generation

- As an overwhelmed student, I want to paste my syllabus so that the AI generates 50 flashcards instantly.
- As a learner, I want to take a dynamically generated quiz to test my knowledge on a specific topic.
- As a reader, I want to summarize a dense 10-page article into bullet points so I can grasp the core concepts quickly.

### Career Preparation

- As a job seeker, I want to upload my resume and target job title so that I know exactly which skills I am missing.
- As a self-taught developer, I want a generated 8-week daily checklist to learn React so I stay on track.

### Gamification

- As a daily user, I want to see my token usage and daily streak so that I am motivated to keep learning.

## 7. Feature Prioritization — MoSCoW

#### Must Have (MVP)

- User Authentication (JWT + OTP).
- Core AI Modules: Neural Chat, Flash Recall, Summarizer.
- Basic Quota Tracking.
- Spaceship / Holographic UI implementation.

#### Should Have (Next Iteration)

- Resume Analyzer module.
- Learning Path Generator with interactive daily tasks.
- Cognitive Quiz module.

### Could Have (Future Enhancements)

- Multi-modal support (Upload images of textbooks).
- Spaced Repetition Algorithm (SM-2) for Flashcards.
- Public user profiles and shareable learning paths.

### Won't Have (For Now)

- Real-time multiplayer study rooms.
- Direct LMS integrations (Canvas, Blackboard).
- Native iOS / Android mobile apps (Focusing on responsive web first).

## 8. Product Roadmap

### Q3 2026: Foundation & MVP Launch

- **Month 1:** Backend infrastructure setup (Spring Boot, MongoDB, Auth).
- **Month 2:** Frontend core UI and integration of Neural Chat, Flashcards, and Summarizer.
- **Month 3:** Beta testing, bug fixes, and V1 Launch.

### Q4 2026: Career Tools & Advanced Learning

- **Month 4:** Release Resume Analyzer and Learning Path Generator.
- **Month 5:** Implement Cognitive Quizzes and advanced token telemetry.
- **Month 6:** Marketing push targeting university students and bootcamp grads.

### Q1 2027: Monetization & Ecosystem Expansion

- **Month 7:** Introduce "Commander" Pro Tier via Stripe ($9.99/mo).
- **Month 8:** Implement Spaced Repetition System (Anki alternative).
- **Month 9:** Release Browser Extension for instant summarization.

## 9. Success Metrics & KPIs

### North Star Metric

- **Active Learning Paths (ALPs):** Number of users completing at least 2 tasks on a Learning Path per week.

### Acquisition Metrics

- **Sign-up Conversion Rate:** Target > 15% from landing page.
- **Customer Acquisition Cost (CAC):** Target < $3.00.

### Engagement Metrics

- **DAU/MAU Ratio:** Target > 30% (App-like daily engagement).
- **Time to First AI Generation:** Target < 2 minutes post-registration.

### Retention Metrics

- **Day 7 Retention Rate:** Target > 40%.
- **Churn Rate on Pro Tier:** Target < 5% monthly.

## 10. Sprint Planning

### Sprint 1: Architecture & Auth (2 Weeks)

- Setup Spring Boot repository, MongoDB schemas, and Groq/Gemini API keys.
- Implement User Registration, OTP via Brevo, and JWT login.
- Scaffold React frontend with Tailwind CSS and routing.

### Sprint 2: Core AI Modules (2 Weeks)

- Develop Prompt Engineering Gateway.
- Implement Neural Chat and Summarizer backend endpoints.
- Build Frontend Chat Interface and Summarizer UI.

### Sprint 3: Flashcards & Quizzes (2 Weeks)

- Implement strict JSON output parsing for Flashcards and Quizzes.
- Build 3D flippable card UI using Framer Motion.

- Develop interactive quiz component with instant grading.

### Sprint 4: Career Tools & Polish (2 Weeks)

- Integrate Apache PDFBox for Resume text extraction.
- Implement Resume Analyzer AI logic and Learning Path Generator.
- Final UI polish, glassmorphism tweaks, and performance optimization.

## 11. Risk Register

| Risk | Impact | Likelihood | Mitigation Strategy |
|---|---|---|---|
| **API Cost Overruns** | High | Medium | Hard-code daily token limits per user; fallback to cheaper LLMs if necessary. |
| **LLM Hallucinations** | High | Low | Strict prompt engineering to enforce fact-based responses; disclaimers in UI. |
| **JSON Parsing Failures** | Medium | Medium | Backend retry logic. Fallback to secondary prompts if initial structure is malformed. |
| **Competitor Cloning** | High | High | Compete heavily on UX/UI and ecosystem lock-in (progress tracking). |

## 12. Architectural Decisions (ADR Log)

- **ADR 001: Use of Groq LLAMA 3.3 70B**
  *Decision:* We chose Groq for its ultra-low latency (800+ tokens/s) to ensure the AI feels instantaneous, bypassing the typical "loading spinner" fatigue of ChatGPT.
- **ADR 002: Stateless Spring Boot Backend**
  *Decision:* No session state is held in memory. Authentication relies purely on JWTs. This allows for horizontal scaling and simple deployments.
- **ADR 003: MongoDB for Schema Flexibility**
  *Decision:* Because AI outputs (especially Learning Paths) are deeply nested JSON objects, a NoSQL database is far more efficient than modeling relational tables.
- **ADR 004: Framer Motion for UI Interactions**
  *Decision:* To achieve the "Spaceship" aesthetic, complex micro-animations are required. Framer Motion provides the necessary declarative API for React.

# 13. Go-To-Market Strategy

### Phase 1: The "Seed" Phase

- **Target Audience:** CS students and self-taught developers.
- **Channels:** Organic TikTok/Reels showcasing the unique UI. Post on Product Hunt and Reddit.
- **Viral Loop:** Users can share their generated Learning Paths on social media via a public link.

### Phase 2: The "Expand" Phase

- **Influencer Partnerships:** Sponsor "Day in the Life" content from tech YouTubers using ArriBot.
- **Programmatic SEO:** Generate thousands of landing pages for specific learning goals (e.g., "How to learn Node.js in 4 weeks").

### Phase 3: Monetization

- Introduce the $9.99/mo Pro Tier offering unlimited generations, the ultra-fast Groq model, and the Resume Analyzer.

# 14. User Journey Map

### Scenario: Preparing for a Tech Interview

1. **Awareness:** User sees a TikTok of ArriBot generating a customized coding curriculum.
2. **Onboarding:** User signs up, verifies OTP, and is greeted by the holographic dashboard.
3. **Action (Resume Analysis):** User navigates to the Resume Analyzer and uploads their PDF resume, typing "Junior Frontend Developer" as the goal.
4. **Insight:** ArriBot highlights that the user is missing "Redux" and "Testing (Jest)" on their resume.
5. **Action (Learning Path):** ArriBot generates a 2-week learning path to learn Redux and Jest.
6. **Execution:** User clicks Day 1 of the path, reads the generated summary, and takes the generated quiz.
7. **Habit Loop:** User returns daily to check off tasks, increasing their system level and keeping their daily streak alive.

## 15. Retrospective — Lessons Learned

*(To be continuously updated during the product lifecycle)*

- **AI Formatting Constraints:** We learned early on that asking an LLM for JSON is not enough. We had to implement robust server-side retry logic and extremely rigid prompt templates to prevent Gson parsing exceptions.
- **The Value of Micro-Interactions:** Initial user testing showed that the platform felt "dead" without animations. Adding Framer Motion drastically increased perceived value and time-on-page.
- **PDF Extraction Challenges:** Extracting text from highly formatted, multi-column PDF resumes is difficult. We had to standardize the input before sending it to the LLM to prevent token waste.
- **Rate Limiting is Mandatory:** Without rate limiting, a single user could exhaust daily API quotas by spamming the generate button. Implementing Guava RateLimiter on the backend was a critical early fix.

***Made by Ishaan Verma***

*"ArriBot — Transforming the way the world learns one neural link at a time."*